Analysis of Post-CHF Heat Transfer for Downward Vertical Boiling in a Square Channel

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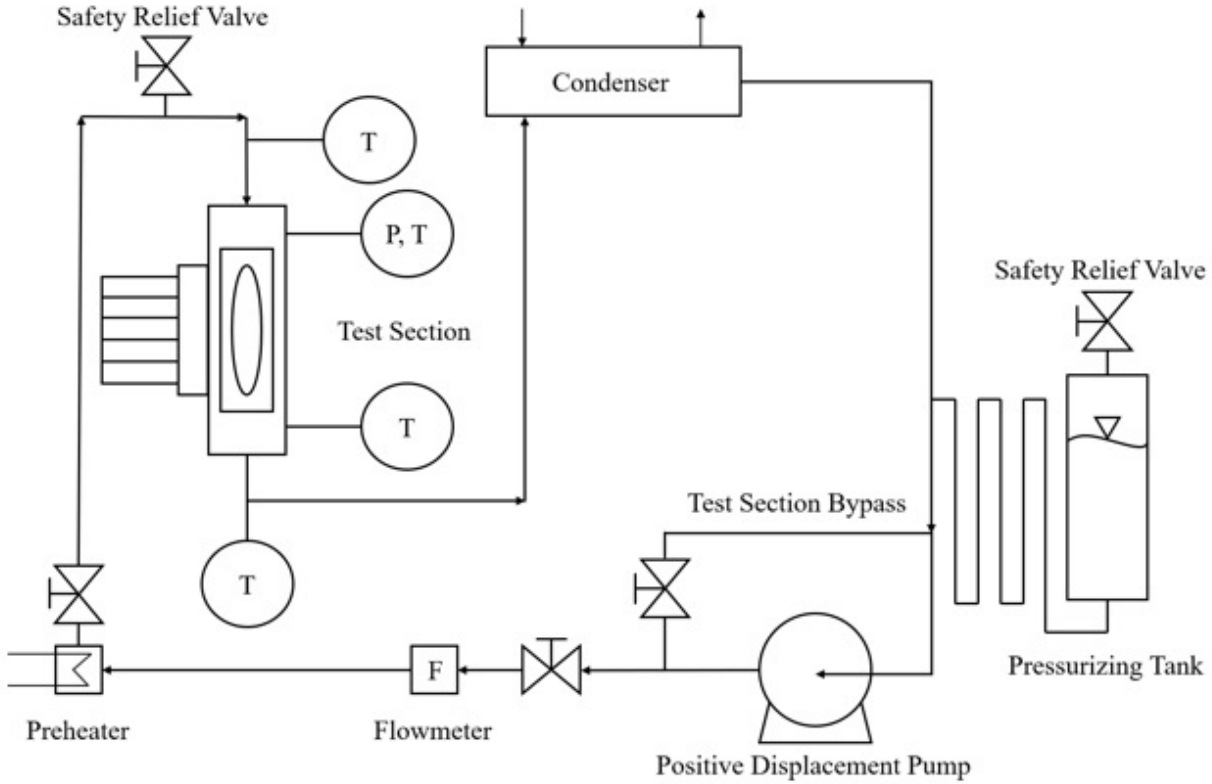
Objectives

1. Streamline a method to measure stable film boiling points and transient cooldown in downwards flow boiling.

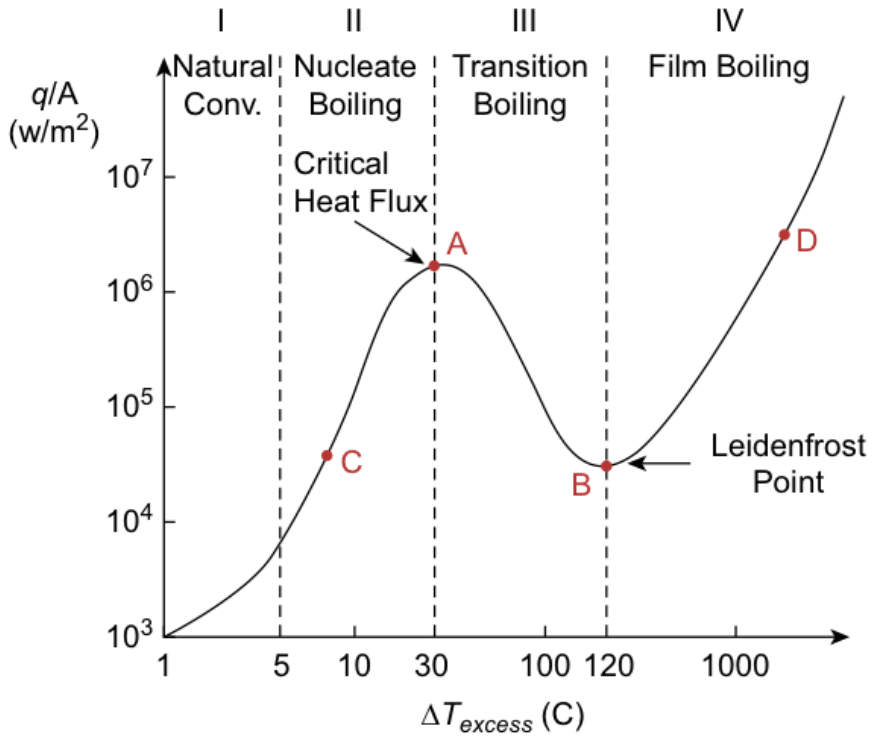
2. Identify characteristic elements of downwards flow in a heated channel and compare with upwards flow of similar conditions.

3. Improve two-phase flow modeling for industrial applications, like nuclear reactors.

Methods



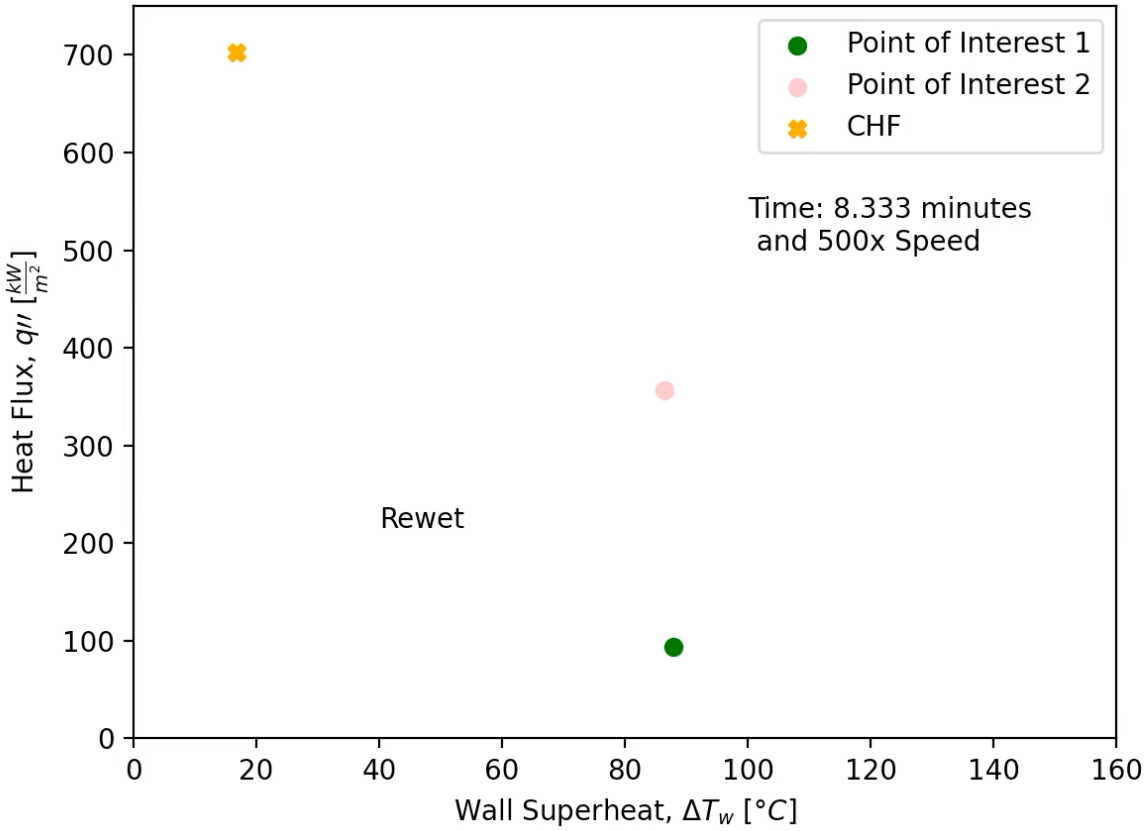
Schematic of the experimental test section [1].



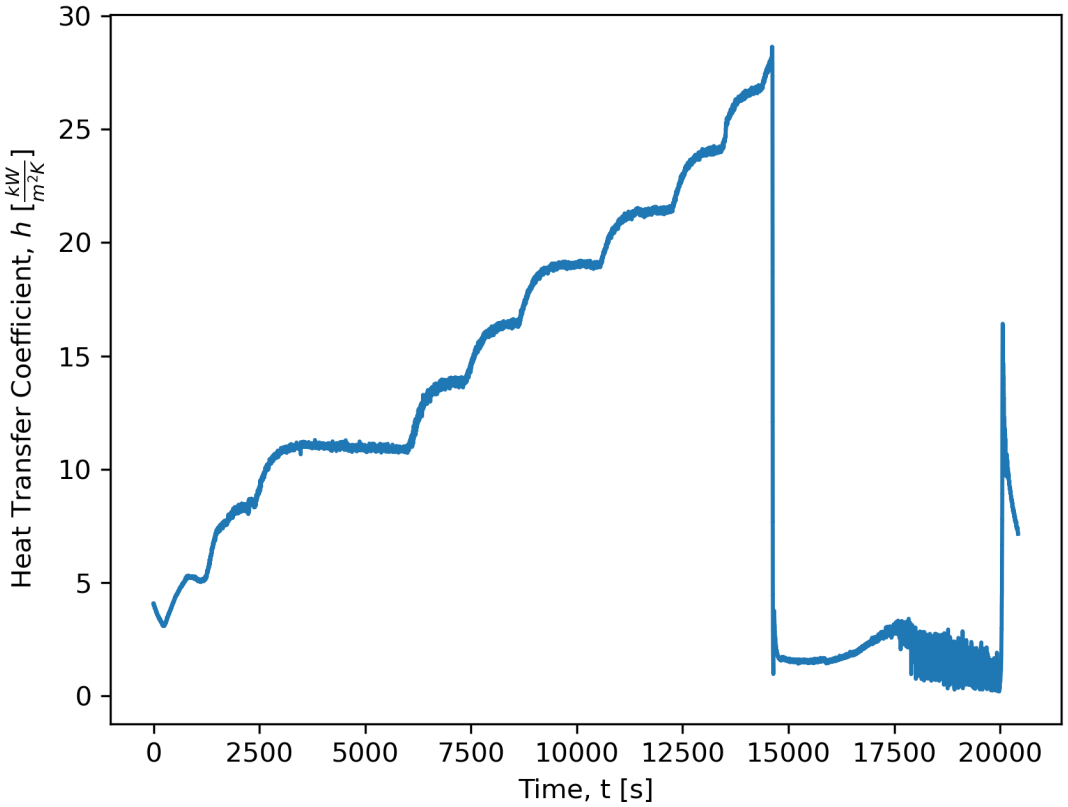
Nukiyama boiling curve [2].

1.0 GPM and 10°C Subcooling Downwards Flow - Results

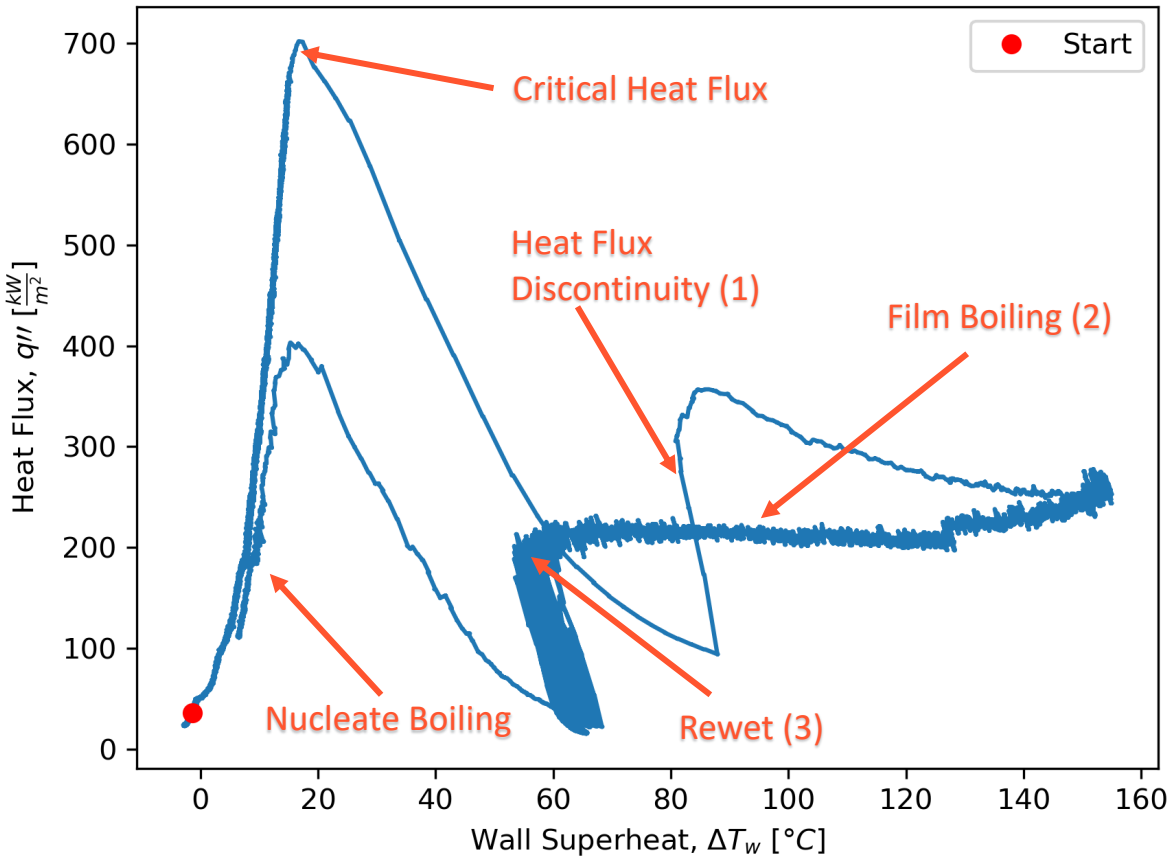
Boiling Curve



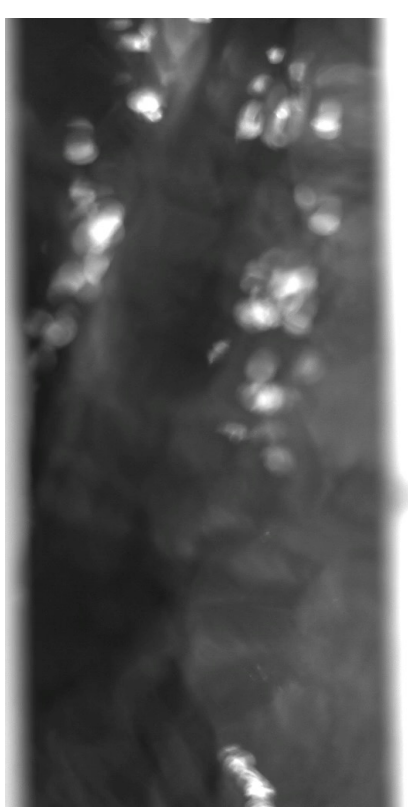
Heat Transfer Coefficient



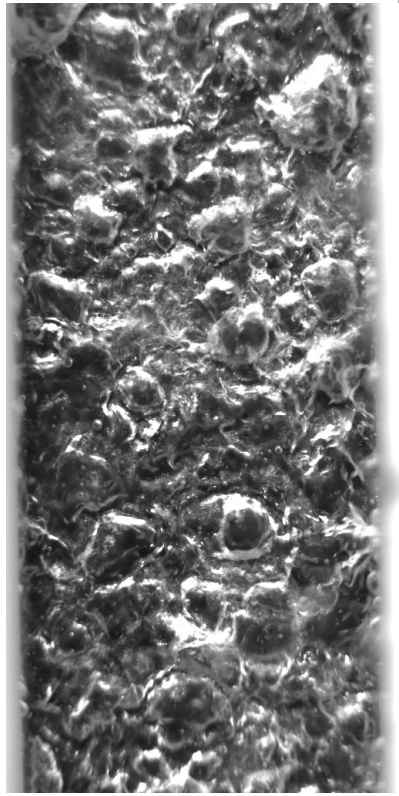
Regional Comparison and Visualization



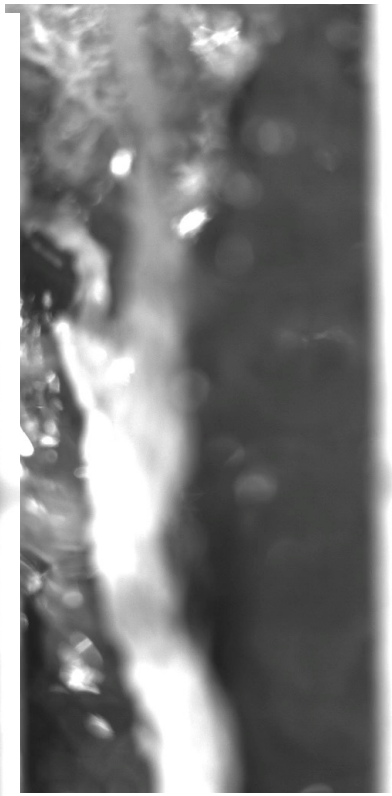
1.0 GPM and 10°C Subcooling



Post-CHF Transient Film Boiling



Steady-State Film Boiling



Rewet Film to Nucleate Boiling

Key Takeaways

1) This is only the beginning.
2) Rewet phenomena vary with condition– further research is necessary to qualify these changes.
3) Understanding downwards flow boiling is critical for safety and efficiency of engineering systems, like nuclear reactors.

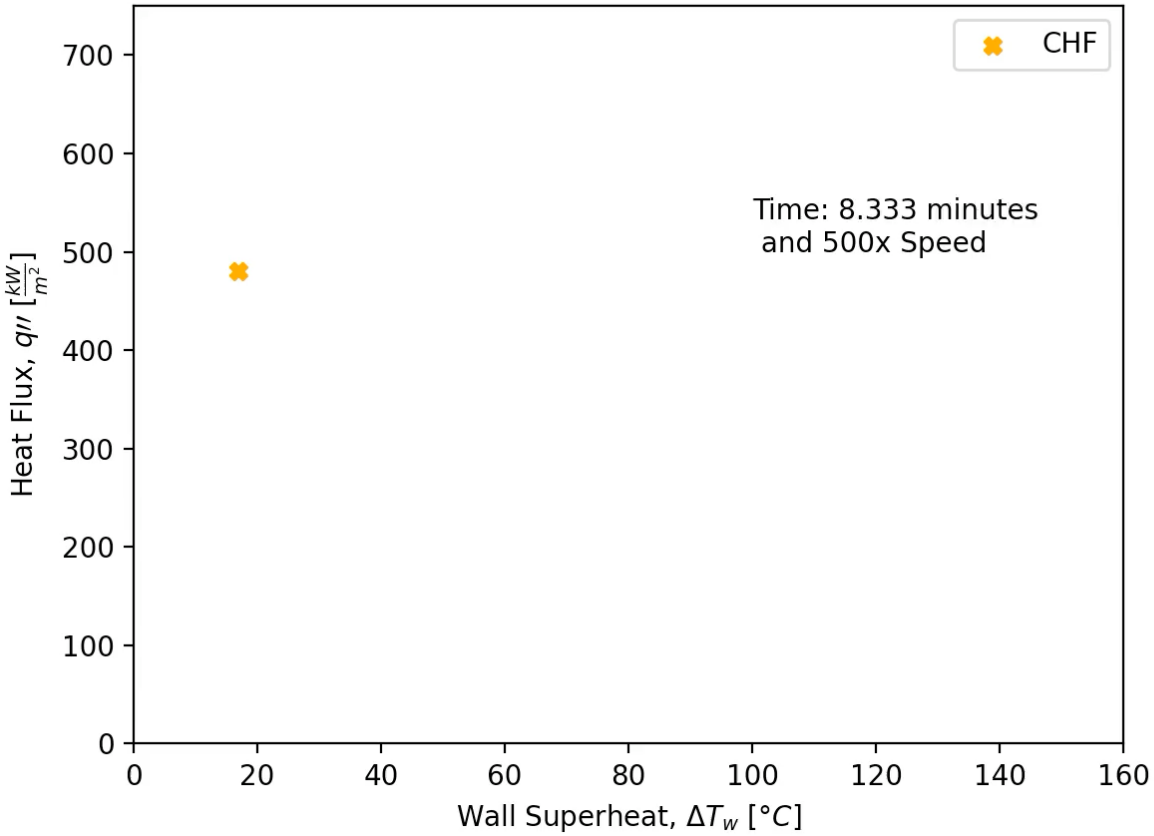
References

[1] M. McCord and C. Brooks, "Experimental comparison of the rewet phenomenon in transient and stable film boiling in low pressure and low flow conditions," International Journal of Heat and Mass Transfer, 2023 (under review).

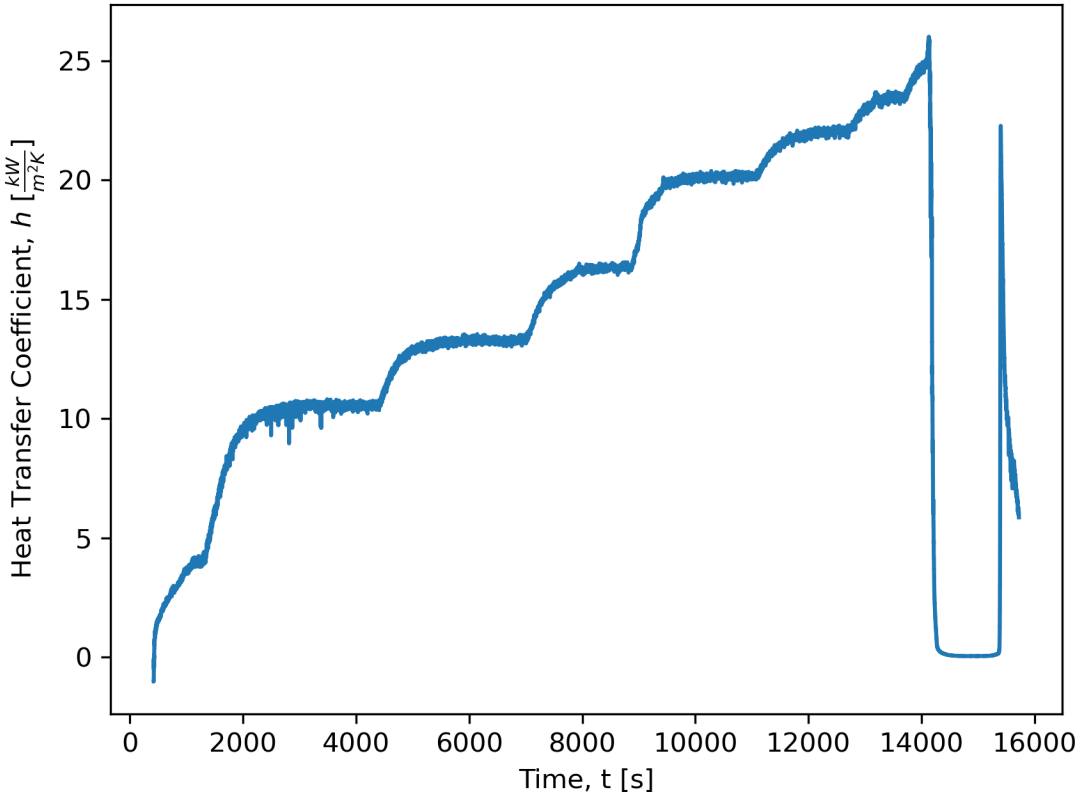
[2] C. H. Forsberg, "Chapter 12 - Special Topics," in *Heat Transfer Principles and Applications*, Academic Press, 2021.

0.5 GPM and 5°C Subcooling Downwards Flow - Results

Boiling Curve



Heat Transfer Coefficient



Additional Results from 1.0 GPM and 10 °C Subcooling

